IT Service Desk Ticket Management System – Project Documentation

1. Project Overview

This project aims to develop an IT Service Desk Ticket Management System that allows organizations to efficiently track, manage, and resolve IT-related issues reported by users. The system provides a structured platform where users can raise support tickets related to hardware, software, network, security, and infrastructure problems.

The system ensures that issues are properly logged, categorized, assigned to appropriate service agents, and resolved in a timely manner. It also provides visibility into ticket status, communication between users and support staff, and historical tracking of issue resolution.

2. Project Objectives

- Provide a centralized system to manage IT service requests.
- Allow users to create and track support tickets.
- Enable administrators to assign tickets to service agents.
- Ensure efficient issue resolution through structured workflows.
- Maintain complete history and status of all tickets.
- Provide dashboards and reports for monitoring service performance.

3. User Roles

- **Customer / User:** Can create support tickets, view ticket status, and confirm resolution.
- **Service Admin:** Manages ticket assignments, monitors ticket progress, and controls system categories.
- **Service Agent:** Works on assigned tickets, resolves issues, and updates ticket status.

4. Ticket Lifecycle Process

- User raises a ticket through the service desk portal.
- Service Admin receives ticket notification.
- Admin reviews and categorizes the issue.
- Ticket is assigned to a Service Agent.
- Agent analyzes the problem.

- Agent fixes the issue if valid.
- Solution is tested and updated in the system.
- User receives resolution notification.
- User confirms the solution.
- Ticket status is marked as closed.

5. System Architecture

The IT Service Desk system follows a three-tier architecture consisting of a frontend interface, a backend API layer, and a database layer.

- **Frontend Layer:** Developed using React.js. It provides the user interface for creating tickets, viewing dashboards, and tracking ticket status.
- **Backend Layer:** Built using ASP.NET Core Web API. This layer handles business logic, authentication, ticket processing, and communication with the database.
- **Database Layer:** SQL Server database used to store users, tickets, categories, comments, attachments, and ticket history.

Overall Architecture Flow: React Frontend → ASP.NET Core API → SQL Server Database